温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Multiple Linear Regression I

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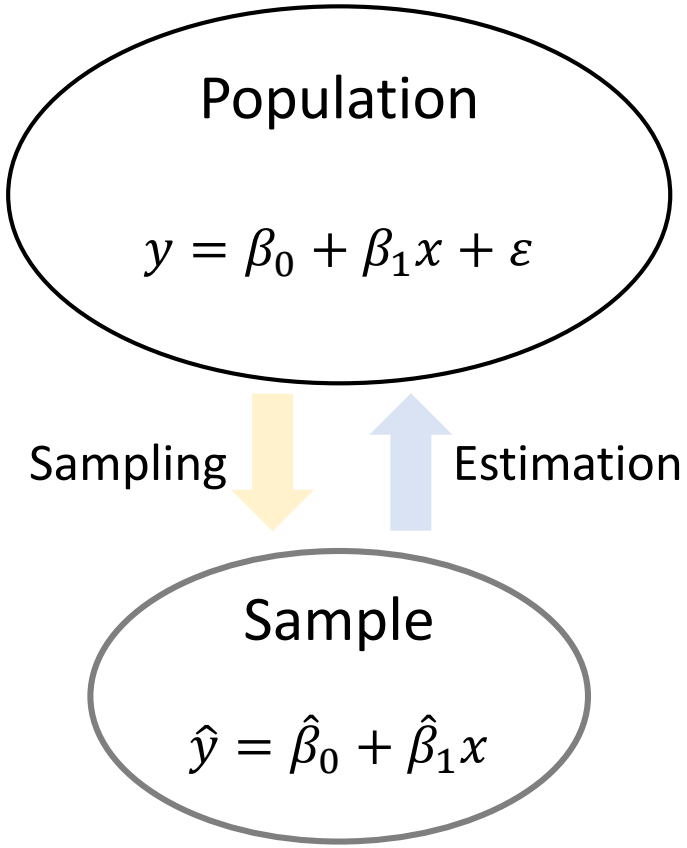
Week 12, Nov 18/20

Review of Simple Linear Regression



OLS estimator: choose $(\hat{\beta}_0, \hat{\beta}_1)$ that minimize the sum of squared residuals

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$$

Gauss-Markov assumptions, properties of $\hat{\beta}_0, \hat{\beta}_1$

Sampling distribution of $\hat{\beta}_0, \hat{\beta}_1$

Hypothesis testing of $\hat{\beta}_0, \hat{\beta}_1$ H_0?

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Review of Simple Linear Regression

```python
import statsmodels.api as sm

x = sm.add_constant(x)
model = sm.OLS(y, x)
results = model.fit()
results.summary()
```

OLS Regression Results

Dep. Variable:	Y	**R-squared:**	0.954
Model:	OLS	**Adj. R-squared:**	0.954
Method:	Least Squares	**F-statistic:**	2046.
Date:	Thu, 13 Nov 2025	**Prob (F-statistic):**	1.79e-67
Time:	18:26:41	**Log-Likelihood:**	-129.13
No. Observations:	100	**AIC:**	262.3
Df Residuals:	98	**BIC:**	267.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	1.9336	0.089	21.723	0.000	1.757	2.110
X	3.6590	0.081	45.234	0.000	3.498	3.820

Omnibus:	5.141	**Durbin-Watson:**	1.763
Prob(Omnibus):	0.076	**Jarque-Bera (JB):**	7.116
Skew:	-0.103	**Prob(JB):**	0.0285
Kurtosis:	4.290	**Cond. No.**	1.11

Interpretation of regression coefficient $\hat{\beta}_0, \hat{\beta}_1$

$$\hat{y} = 1.93 + 3.66x$$

Motivation: Many IVs

- Why simple linear regression is not enough
  - Real outcomes are influenced by more than one factor.
  - Ignoring important variables → **biased** or **misleading** results/relationships.
  - Example: GPA is influenced by study hours, course load, ability, etc.
- Key idea
  - To isolate the effect of one variable, we must hold others factors constant.
- Transition
  - Multiple linear regression analyzes

$$y \sim x_1 + x_2 + x_3 + \ \ldots + x_k$$

→ moving from a line (1 IV) to a plane/surface (many IVs)

Multiple Linear Regression

Population $\quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k\, x_k + \varepsilon$

Sample $\quad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \cdots + \hat{\beta}_k x_k$

Vector Form $\quad \underbrace{\boldsymbol{y}}_{n\times 1} = \underbrace{\boldsymbol{X}}_{n\times(k+1)} \underbrace{\boldsymbol{\beta}}_{(k+1)\times 1} + \underbrace{\boldsymbol{\varepsilon}}_{n\times 1}$

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \boldsymbol{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Estimation – OLS Estimator

- Goal
  - Choose $(\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k)$ to minimize the sum of the squared errors between the observed value y_i and the predicted value $\hat{y}_i$

$$\boldsymbol{min \sum (y_i - \hat{y}_i)^2 = min \sum \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k\right)\right)^2 = min \sum e_i^2}$$

- Computation Note
  - Because the optimization involves the use of **matrix algebra**, we rely on computer software to compute the estimates.
  - Our focus will be on understanding and interpreting the output – not deriving the formulas.

Interpretation

Sample $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \cdots + \hat{\beta}_k x_k$

Coefficient on x_1 /
Partial coefficient on x_1

- The partial regression coefficients represent the expected change in DV when the associated IV is increased by **one unit** while the values of all other IVs are held constant.

I'll say it again:

- Each coefficient $\hat{\beta}_i$ estimates the mean change in the DV(y) **per unit** increase in x_i, when all other predictors are held constant.

Interpretation

$$\textbf{Profit} = -2000 + 2.5 * \textbf{ExpenditureOnAdvertising} + 32 * \textbf{NumberOfProductsSold}$$

- $\hat{\beta}_0$: Monthly profit is $-\$2000$ <u>*without any money spent on advertising and with zero products sold*</u>. $\leftarrow$ Reasonable?
- $\hat{\beta}_1$: Holding the number of products sold constant, every dollar spent on advertising increases profit by $\$2.50$.
- $\hat{\beta}_2$: Keeping advertising expenditure constant, every product sold increases profit by $\$32$.

Interpretation – Categorical IVs

- So far we have been dealing with continuous IVs (X) (e.g. expenditure, years, age, numbers, …)
- Next, we consider categorical IVs (e.g., gender, ethnicity, marital status, major, …).

Example:

$$\textbf{UmbrellaSales} = \hat{\beta}_0 + \hat{\beta}_1 \textbf{Weather}$$

- These categorical variables take one of a small set of fixed values.
- For example, let's assume that:

Weather can be either "**Sunny**" or "**Rainy**".

Interpretation – Categorical IVs

- Dummy Coding is a method by which we create and use "dummy variables" in our regression model.

In this example, we can define a variable: "Rainy"

that is 1 if "Weather==Rainy", and 0 if "Weather==Sunny"

"Rainy" is called a dummy variable

$$\textbf{UmbrellaSales} = \hat{\beta}_0 + \hat{\beta}_1 \textbf{Weather} \quad \Rightarrow \quad \textbf{UmbrellaSales} = \hat{\beta}_0 + \hat{\beta}_1 \textbf{Rainy}$$

Thus, this breaks down into two equations:

- If Sunny, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(0) = \hat{\beta}_0$
- If Rainy, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(1) = \hat{\beta}_0 + \hat{\beta}_1$

Interpretation – Categorical IVs

$$\textbf{UmbrellaSales} = \hat{\beta}_0 + \hat{\beta}_1 \textbf{Rainy}$$

If Sunny, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(0) = \hat{\beta}_0$

If Rainy, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(1) = \hat{\beta}_0 + \hat{\beta}_1$

<u>Interpretation:</u>

- $\hat{\beta}_0$: **Average** Umbrella sales when it is Sunny
- $\hat{\beta}_0 + \hat{\beta}_1$: **Average** Umbrella sales when it is Rainy
- $\hat{\beta}_1$: **Average difference** in umbrella sales when it is Rainy, *compared to when it is Sunny* (Sales when Rainy - Sales when Sunny)

- Q: Would you predict $\hat{\beta}_1$ to be greater than 0 or less than $\hat{\beta}_0$?

Interpretation – Categorical IVs

- If, instead, Weather can now take one of three values:
  { “Sunny”, “Rainy”, “Cloudy” }
- Then we can define two dummy variables, Rainy and Cloudy,
  Rainy = 1 if weather is rainy, 0 otherwise
  Cloudy = 1 if weather is cloudy, 0 otherwise
- We say that “Sunny” is the Reference Group for the categorical variable “Weather”.

$$\textbf{UmbrellaSales} = \hat{\beta}_0 + \hat{\beta}_1 \textbf{Rainy} + \hat{\beta}_2 \textbf{Cloudy}$$

If Sunny, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(0) + \hat{\beta}_2(0) = \hat{\beta}_0$

If Rainy, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(1) + \hat{\beta}_2(0) = \hat{\beta}_0 + \hat{\beta}_1$

If Cloudy, **UmbrellaSales** = $\hat{\beta}_0 + \hat{\beta}_1(0) + \hat{\beta}_2(1) = \hat{\beta}_0 + \hat{\beta}_2$

Interpretation – Categorical IVs

- More generally, a categorical variable with n levels will have $(n-1)$ dummy variables.
- $\hat{\beta}_0$: Average **Value** of y for the reference group.
- $\hat{\beta}_i$: Average **Difference** in y for Dummy Group i compared to the reference group.

- The choice of reference group is not fixed. And choosing your reference group well (depending on your goals) will make your analyses more convenient and interpretable.
- For example, for this Umbrella Sales example ({ “Sunny”, “Rainy”, “Cloudy” }),
- I think “Sunny” is a good reference group. Why?

Current Button | Button A | Button B | Button C

Which button is a good reference group?

Interpretation – Categorical IVs

Whenever we have categorical independent variables in a model, interpreting the coefficients has to be done *with respect to the reference group*.

Example:

Umbrella Sales is a continuous variable measured in $.

Rainy and Cloudy are dummy variables ("Sunny" is the reference group)

ExpenditureOnAdvertising is also a continuous variable measured in $.

$$\textbf{UmbrellaSales} = 10 + 50*\textbf{Rainy} + 20*\textbf{Cloudy} + 2.5*\textbf{ExpenditureOnAdvertising}$$

Interpret the output:

- 10: Average umbrella sales when _____ and _____ .
- 50: Average umbrella sales when _____ compared to_____, and_____.
- 20: Average umbrella sales when _____ compared to_____, and_____ .
- 2.5: When_____, every dollar spent on advertising _____.

Compare Coefficients: Simple vs. Multiple Regression

Simple Linear Regression $y = \beta_0 + \beta_1^{SLR} x_1 + \varepsilon$

Multiple Linear Regression $y = \beta_0 + \beta_1^{MLR} x_1 + \beta_2 x_2 + \cdots + \varepsilon$

$$\beta_1^{SLR} = \beta_1^{MLR}?$$

SLR: $y = 2.0 + 0.06x_1$

MLR: $y = 1.8 + 0.03x_1 + 0.40x_2$

y is GPA, x_1 is study hour, x_2 is attendance, correlation between study hours and attendance is 0.7

Takeaway

- SLR coefficient: effect of x_1 *by itself*
- MLR coefficient: effect of x_1 *after removing the influence of other variables*

Multicollinearity

- What is multicollinearity (多重共线性)?
  - When two or more IVs are highly correlated.
  - The predictors contain overlapping information about the DV.
- Why it is a problem
  - Coefficients become **unstable** and sensitive to small data changes.
  - Standard errors become **large** → coefficients may appear insignificant.
  - Signs of coefficients may become **counterintuitive** or flip unexpectedly.
  - Hard to assess each variable's **partial effect**.

Multicollinearity

- How to detect it
  - Correlation matrix: look for high correlations among IVs
  - Variance Inflation Factor (VIF) 方差膨胀因子

    Rule of thumb: **VIF > 5 (or 10)** indicates strong multicollinearity.

$$x_1 = \alpha_0 + \alpha_2 x_2 + \alpha_3 x_3 + \cdots + \alpha_k x_k + \varepsilon$$

The VIF factor for $\hat{\alpha}_i$ is: $VIF_i = \frac{1}{1-R_i^2}$

How to compute correlation matrix, VIF in Python?

What does VIF=1 mean?

- What to do
  - Remove one of the correlated variables
  - Combine variables (e.g., ratios, indices)
  - Use domain knowledge to select the more relevant IV

Adjusted $\boldsymbol{R^2}$

- Why adjust R^2? R^2 reward complexity. Why?
  - In multiple regression, adding more predictors **always increases** R^2, even if the new variables are useless.
  - We need a measure that rewards **useful predictors** and penalizes **noise**.
- Definition

$$R_{adj}^2 = 1 - \left(\frac{SSE/(n-k-1)}{SST/(n-1)}\right)$$

  where:
  - n: sample size
  - k: number of predictors / IVs
- Key Properties
  - $R_{adj}^2 \leq R^2$
  - Can decrease when adding a weak predictor, increase only if the new IV improves the model more than expected by chance

Test Your Understanding

1. For this simple linear regression,

$$\text{Income} = \hat{\beta}_0 + \hat{\beta}_1 * \text{Years of Education}$$

We find that $\hat{\beta}_0 = -4500$, $\hat{\beta}_1 = 500$ (in \$/year). What do these mean? [Negative income?!]

2. For this multiple linear regression,

$$\text{Income} = \hat{\beta}_0 + \hat{\beta}_1 * \text{Years of Education} + \hat{\beta}_2 * \text{Years of Experience}$$

We find $\hat{\beta}_0 = -3700$, $\hat{\beta}_1 = 450$, $\hat{\beta}_2 = 350$ (in \$/year). What do these mean?

3. For this simple linear regression,

$$\text{Income} = \hat{\beta}_0 + \hat{\beta}_1 * \text{Gender}$$

Let us say Gender = 1 if Female, 0 if Male.

We find that $\hat{\beta}_0 = 3500$, $\hat{\beta}_1 = 5$ (in \$/year). What does this mean?

Summary

- **Multiple Linear Regression:** estimates the partial effect of each predictor while holding all others constant.
- **Coefficient Interpretation:** understand the direction, size, and statistical significance of each IV (continuous, categorical).
- **SLR vs. MLR:** coefficients may differ due to omitted variable bias when important predictors are left out.
- **Multicollinearity:** identify using the correlation matrix and VIF; it inflates standard errors and destabilizes estimates.
- **Adjusted R^2:** measures model fit while penalizing unnecessary predictors.

Remaining Course Schedule

W12	11/18 11/20	Multiple Linear Regression I
W13	11/25 11/27	Multiple Linear Regression II
W14	12/02 12/04	Time Series Analysis
W15	12/09 12/11	Q&A for Group Project Group Project Presentation
W16	12/16 12/18	Review, Q&A for Final Exam Final Exam (written, closed book)

Assessment

Grade Breakdown	
Participation	10%
Assignments	3x5%=15%
Midterm Exam	25%
Group Project	25%
Final Exam	25%
Total	100%

Participation: Class participant is mandatory according to school policy.
1 point will be deducted for each class absence without approval.

Adjusted Office Hours

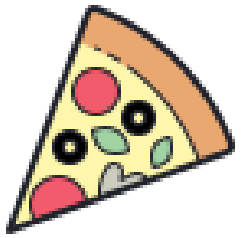
Monday	11:00 – 12:30
Wednesday	11:00 – 12:30
Friday	15:00 – 17:00

Group Projects Grouping

Group 1
Kexin Zhang
Siyuan Chen (Crissy)
Zhien Yu (June)
Lixian Fan (JUNO)

Group 2
Ye Guan
Chi Xu (Rosalyn)
Xinling Chen (Cyndi)
Zhehao Shen (Steven)

Group 3
Chengkang Zhang (Keith)
Mingyang Liu (Jack)
Xiaoxue He (Eirwen)
Rui Wang (Bruce)



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